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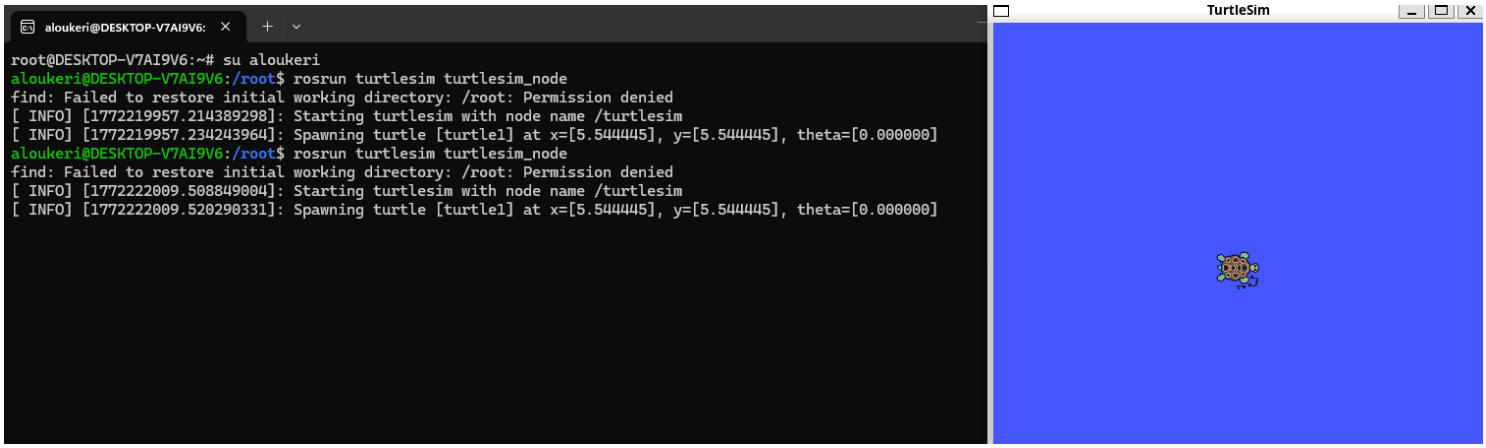
Robotics' Control - School of Informatics
Assignment 02.

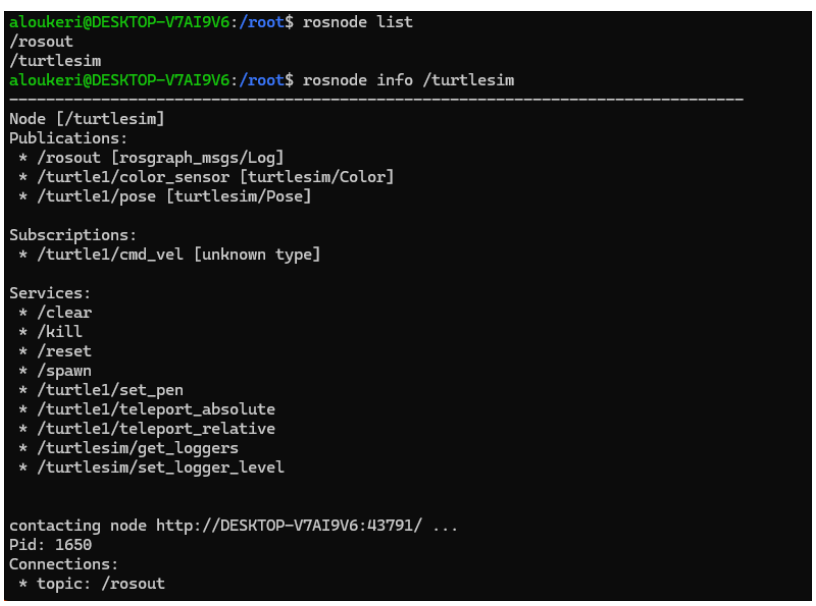
1) Initial Setup

Initializing the ROS Master by running roscore and launching the turtlesim_node to display the simulation environment. Confirmation of the node starting with name /turtlesim and spawning the turtle at specific coordinates ($x = 5.54$, $y = 5.54$, $\theta = 0$).



2) Node Exploration

Utilizing rosnode list to identify active nodes, specifically /turtlesim and /rosout. Documenting node details via rosnode info, including its published topics like /turtle1/pose and subscriptions like /turtle1/cmd_vel.



3) Topic Identification

Creating a comprehensive list of all topics associated with the turtlesim node using rostopic list. Mapping each topic to its specific message type, such as geometry_msgs/Twist for velocity commands and turtlesim/Pose for position data.

```
contacting node http://DESKTOP-V7AI9V6:43791/ ...
Pid: 1650
Connections:
 * topic: /rosout
    * to: /rosout
    * direction: outbound (37353 - 127.0.0.1:54082) [16]
    * transport: TCPROS

aloukeri@DESKTOP-V7AI9V6:/root$ rostopic list
/rosout
/rosout_agg
/turtle1/cmd_vel
/turtle1/color_sensor
/turtle1/pose
aloukeri@DESKTOP-V7AI9V6:/root$ rostopic type /rosout
rosgraph_msgs/Log
aloukeri@DESKTOP-V7AI9V6:/root$ rostopic type /rosout_agg
rosgraph_msgs/Log
aloukeri@DESKTOP-V7AI9V6:/root$ rostopic type turtle1/cmd_vel
geometry_msgs/Twist
aloukeri@DESKTOP-V7AI9V6:/root$ rostopic type turtle1/color_sensor
turtlesim/Color
aloukeri@DESKTOP-V7AI9V6:/root$ rostopic type turtle1/pose
turtlesim/Pose
aloukeri@DESKTOP-V7AI9V6:/root$
```

4a) Message Types

```
aloukeri@DESKTOP-V7AI9V6:/root$ rosmsg list
actionlib/TestAction
actionlib/TestActionFeedback
actionlib/TestActionGoal
actionlib/TestActionResult
actionlib/TestFeedback
actionlib/TestGoal
actionlib/TestRequestAction
actionlib/TestRequestActionFeedback
actionlib/TestRequestActionGoal
actionlib/TestRequestActionResult
actionlib/TestRequestFeedback
actionlib/TestRequestGoal
actionlib/TestRequestResult
actionlib/TestResult
actionlib/TwoIntsAction
actionlib/TwoIntsActionFeedback
actionlib/TwoIntsActionGoal
actionlib/TwoIntsActionResult
actionlib/TwoIntsFeedback
actionlib/TwoIntsGoal
actionlib/TwoIntsResult
actionlib_msgs/GoalID
actionlib_msgs/GoalStatus
actionlib_msgs/GoalStatusArray
actionlib_tutorials/AveragingAction
actionlib_tutorials/AveragingActionFeedback
actionlib_tutorials/AveragingActionGoal
```

```
actionlib_tutorials/AveragingActionFeedback
actionlib_tutorials/AveragingActionGoal
actionlib_tutorials/AveragingActionResult
actionlib_tutorials/AveragingFeedback
actionlib_tutorials/AveragingGoal
actionlib_tutorials/AveragingResult
actionlib_tutorials/FibonacciAction
actionlib_tutorials/FibonacciActionFeedback
actionlib_tutorials/FibonacciActionGoal
actionlib_tutorials/FibonacciActionResult
actionlib_tutorials/FibonacciFeedback
actionlib_tutorials/FibonacciGoal
actionlib_tutorials/FibonacciResult
bond/Constants
bond/Status
control_msgs/FollowJointTrajectoryAction
control_msgs/FollowJointTrajectoryActionFeedback
control_msgs/FollowJointTrajectoryActionGoal
control_msgs/FollowJointTrajectoryActionResult
control_msgs/FollowJointTrajectoryFeedback
control_msgs/FollowJointTrajectoryGoal
control_msgs/FollowJointTrajectoryResult
control_msgs/GripperCommand
control_msgs/GripperCommandAction
control_msgs/GripperCommandActionFeedback
control_msgs/GripperCommandActionGoal
control_msgs/GripperCommandActionResult
control_msgs/GripperCommandFeedback
control_msgs/GripperCommandGoal
control_msgs/GripperCommandResult
```

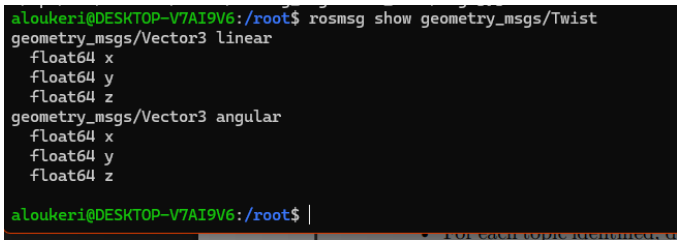
```
control_msgs/GripperCommandResult
control_msgs/JointControllerState
control_msgs/JointJog
control_msgs/JointTolerance
control_msgs/JointTrajectoryAction
control_msgs/JointTrajectoryActionFeedback
control_msgs/JointTrajectoryActionGoal
control_msgs/JointTrajectoryActionResult
control_msgs/JointTrajectoryControllerState
control_msgs/JointTrajectoryFeedback
control_msgs/JointTrajectoryGoal
control_msgs/JointTrajectoryResult
control_msgs/PidState
control_msgs/PointHeadAction
control_msgs/PointHeadActionFeedback
control_msgs/PointHeadActionGoal
control_msgs/PointHeadActionResult
control_msgs/PointHeadFeedback
control_msgs/PointHeadGoal
control_msgs/PointHeadResult
control_msgs/SingleJointPositionAction
control_msgs/SingleJointPositionActionFeedback
control_msgs/SingleJointPositionActionGoal
control_msgs/SingleJointPositionActionResult
control_msgs/SingleJointPositionFeedback
control_msgs/SingleJointPositionGoal
control_msgs/SingleJointPositionResult
controller_manager_msgs/ControllerState
controller_manager_msgs/ControllerStatistics
controller_manager_msgs/ControllersStatistics
```

```
controller_manager_msgs/HardwareInterfaceResources
diagnostic_msgs/DiagnosticArray
diagnostic_msgs/DiagnosticStatus
diagnostic_msgs/KeyValue
dynamic_reconfigure/BoolParameter
dynamic_reconfigure/Config
dynamic_reconfigure/ConfigDescription
dynamic_reconfigure/DoubleParameter
dynamic_reconfigure/Group
dynamic_reconfigure/GroupState
dynamic_reconfigure/IntParameter
dynamic_reconfigure/ParamDescription
dynamic_reconfigure/SensorLevels
dynamic_reconfigure/StrParameter
gazebo_msgs/ContactState
gazebo_msgs/ContactsState
gazebo_msgs/LinkState
gazebo_msgs/LinkStates
gazebo_msgs/ModelState
gazebo_msgs/ModelStates
gazebo_msgs/ODEJointProperties
gazebo_msgs/ODEPhysics
gazebo_msgs/WorldState
geometry_msgs/Accel
geometry_msgs/AccelStamped
geometry_msgs/AccelWithCovariance
geometry_msgs/AccelWithCovarianceStamped
geometry_msgs/Inertia
geometry_msgs/InertiaStamped
geometry_msgs/Point
```

4b)

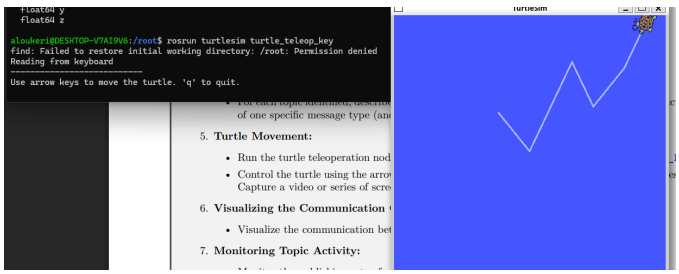
```
aloukeri@DESKTOP-V7AI9V6:/root$ rosmsg show gazebo_msgs/ContactState
string info
string collision1_name
string collision2_name
geometry_msgs/Wrench[] wrenches
  geometry_msgs/Vector3 force
    float64 x
    float64 y
    float64 z
  geometry_msgs/Vector3 torque
    float64 x
    float64 y
    float64 z
geometry_msgs/Wrench total_wrench
  geometry_msgs/Vector3 force
    float64 x
    float64 y
    float64 z
  geometry_msgs/Vector3 torque
    float64 x
    float64 y
    float64 z
geometry_msgs/Vector3[] contact_positions
  float64 x
  float64 y
  float64 z
geometry_msgs/Vector3[] contact_normals
  float64 x
  float64 y
  float64 z
```

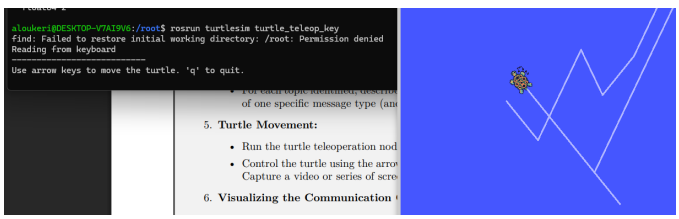
4c) This is a second approach because the question confused me a little. It refers to topic turtle1/cmd_vel . Investigating the internal structure of the messages used by the topics. Using rosmsg show to display the data fields of geometry_msgs/Twist, which consists of linear and angular vectors, each containing x, y, z coordinates of type float64.



5) Turtle Movement

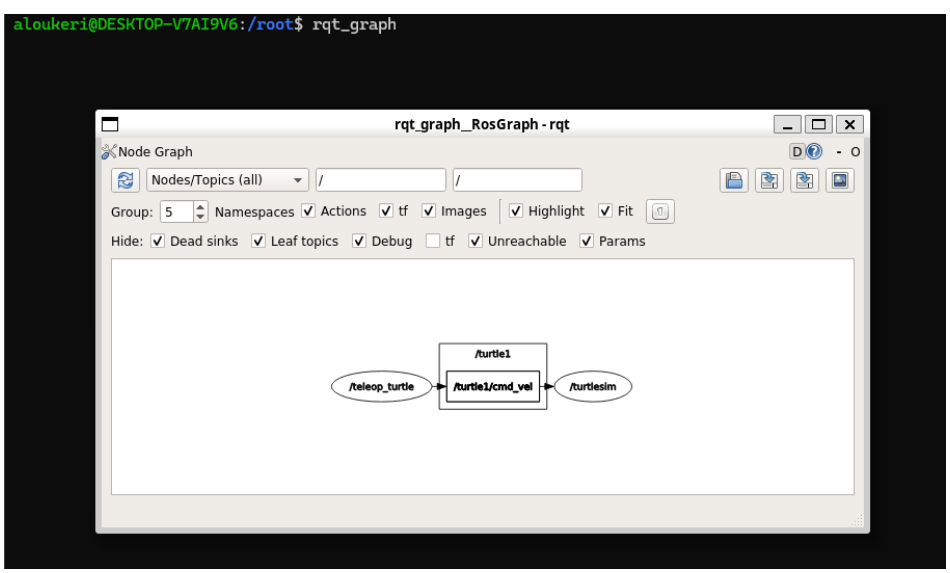
Executing the turtle_teleop_key node to enable manual control of the robot. Using the arrow keys to send movement commands, resulting in the turtle drawing paths within the graphic interface.





6) Communication Graph

The graph confirms the communication flow where the /teleop_turtle node publishes messages to the /turtlesim node via the /turtle1/cmd_vel topic.



7) Monitoring Topic Activity

Recording the publishing rate in Hz using rostopic hz and measuring the data bandwidth in Bytes per second using rostopic bw.

```
aloukeri@DESKTOP-V7AI9V6:/root$ rostopic hz turtle1/cmd_vel
subscribed to [/turtle1/cmd_vel]
no new messages
no new messages
no new messages
no new messages
no new messages
no new messages
no new messages
no new messages
no new messages
no new messages
no new messages
no new messages
average rate: 1.654
        min: 0.604s max: 0.604s std dev: 0.00000s window: 2
average rate: 1.241
        min: 0.604s max: 1.007s std dev: 0.20109s window: 3
no new messages
no new messages
average rate: 0.855
        min: 0.338s max: 2.731s std dev: 0.93226s window: 5
average rate: 0.974
        min: 0.338s max: 2.731s std dev: 0.88185s window: 6
average rate: 1.000
        min: 0.338s max: 2.731s std dev: 0.80727s window: 7
no new messages
no new messages
```

```
aloukeri@DESKTOP-V7AI9V6:/root$ rostopic bw turtle1/cmd_vel
subscribed to [/turtle1/cmd_vel]
average: 82.67B/s
        mean: 48.00B min: 48.00B max: 48.00B window: 3
average: 87.48B/s
        mean: 48.00B min: 48.00B max: 48.00B window: 5
average: 102.53B/s
        mean: 48.00B min: 48.00B max: 48.00B window: 8
average: 131.45B/s
        mean: 48.00B min: 48.00B max: 48.00B window: 13
average: 108.55B/s
        mean: 48.00B min: 48.00B max: 48.00B window: 13
average: 92.44B/s
        mean: 48.00B min: 48.00B max: 48.00B window: 13
average: 80.50B/s
        mean: 48.00B min: 48.00B max: 48.00B window: 13
```